BRADLEY
University

CS 330: Net-Centric Computing – Spring 2024

Professor: Dr. Mohammad N. Sadat

Homework 2

- Cheating or plagiarism will result in a grade of zero for the entire assessment.
- Required screenshots must be included as evidence to receive full credit.
- Skeleton code for the programming questions can be downloaded from Canvas.

Question:	1	2	3	Total
Points:	6	6	3	15

Q.1 Socket Programming: Web Server

You will develop a simple Web server that is capable of processing only one request. Specifically, your Web server will

1. create a connection socket when contacted by a client (browser),
2. receive the HTTP request from this connection,
3. parse the request to determine the specific file being requested,
4. get the requested file from the server's file system,
5. create an HTTP response message consisting of the requested file preceded by header lines, and
6. send the response over the TCP connection to the requesting browser.

If a browser requests a file that is not present in your server, your server should return a "404 Not Found" error message.

Code: You can find a skeleton code (in Python) for your server on Canvas. The places where you need to fill in code are marked with `#Fill in start` and `#Fill in end`. Each place may require one or more lines of code. You are allowed to use a different programming language, however, the skeleton code is only available in Python.

Your job is to complete the code, run your server, and then test your server by sending requests from browsers running on different hosts. If you run your server on a host that already has a Web server running on it, then you should use a different port than port 80 for your Web server.

Running the Server: Put an HTML file (e.g., `HelloWorld.html`) in the same directory that the server is in. If you are not familiar with HTML, you can find sample HTML code here: [1], [2], or [3]. Run the server program. Determine the IP address of the host that is running the server (e.g., `128.238.251.26`). From another host, open a browser and provide the corresponding URL. For example:

`http://128.238.251.26:6789/HelloWorld.html`

In the URL above, `HelloWorld.html` is the name of the file you placed in the server directory. Note also the use of the port number after the colon. You need to replace this port number with whatever port you have used in the server code. In the above example, we have used the port number `6789`. The browser should then display the contents of `HelloWorld.html`. If you omit "`:6789`", the browser will assume port `80` and you will get the web page from the server only if your server is listening at port `80`. Then try to get a file that is not present at the server. You should get a "404 Not Found" message.

Submit the following for this task:

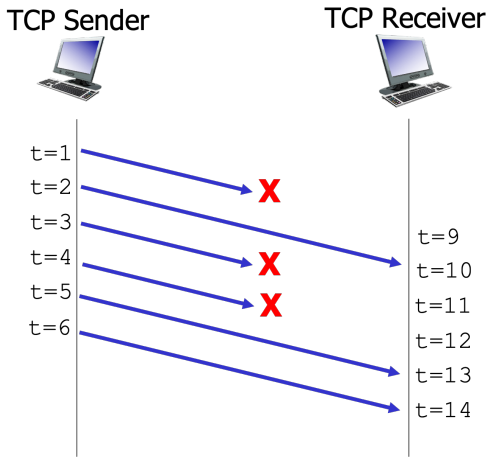
(a) One (1) .py file containing the complete Python code for the server program. [2]

(b) One (1) .html file containing the HTML code for the webpage. [1]

(c) Screenshot(s) of your client browser, verifying that you received the contents of the HTML file from the server. When you put the screenshot in your Word file, properly crop and resize it to ensure good readability. [3]

Q.2 Consider the figure below in which a TCP sender and receiver communicate over a connection in which the sender→receiver segments may be lost. The TCP sender sends an initial window of six (6) segments. As shown in the figure below, some of the segments are lost between the sender and the receiver.

Suppose the initial value of the sender→receiver sequence number is 350, and the segments each contain 890 bytes. The delay between the sender and receiver is 8 time units.



(a) What are the sequence numbers associated with each of the six segments sent by the sender? Format your answer as: a, b, c, ... [3]

(b) Determine the ACK numbers the receiver sends in response to each of the segments. If a segment never arrives, use 'X' to denote it. Format your answer as: a, b, c, ... [3]

Q.3 Compute the Internet checksum value for the following two 16-bit words: [3]

`01000001 11000100`

and

`00100010 00101011`

Your answer must be formatted to have a single space between the two octets, e.g., `xxxxxxxx yyyyyyyy`.

Grading Rubric

Question #	Criteria	Weight
1	1. Python code (.py file) is correct and included with the submission. The HTML file for the webpage (.html) is also included.	50%
	2. All required screenshots are submitted, showing all the required steps listed in the question.	50%
2 - 3	1. All components of the question have been addressed correctly.	90%
	2. The answers are formatted as directed.	10%

Final Report Submission on Canvas

- ❑ **Files to upload:** Upload one (1) PDF file + one (1) Python file + one (1) HTML file. Do NOT compress/ZIP.
  - ☞ **PDF:** Include all essay question answers, AND all screenshots for the programming questions with appropriate captions/descriptions in the same PDF.
  - ☞ **Python and HTML files:** The Python and HTML files should be uploaded directly to Canvas.